

SOC ANALYST INVESTIGATION REPORT

EventID: 82 | Rule: SOC140 – Phishing Mail Detected – Suspicious Task Scheduler
Analyst: Mehenni May | Platform: LetsDefend | Date: Feb 23, 2026

 **VERDICT: TRUE POSITIVE — PHISHING + MALWARE DELIVERY** 

PHASE 0 — ALERT OVERVIEW

EventID	82
Event Time	March 21, 2021, 12:26 PM
Rule Triggered	SOC140 – Phishing Mail Detected – Suspicious Task Scheduler
Analyst Level	Security Analyst
Device Action	Blocked ✓
Final Verdict	TRUE POSITIVE — Phishing + Malware

PHASE 1 — EMAIL & INFRASTRUCTURE ANALYSIS

1.1 Email Header Analysis

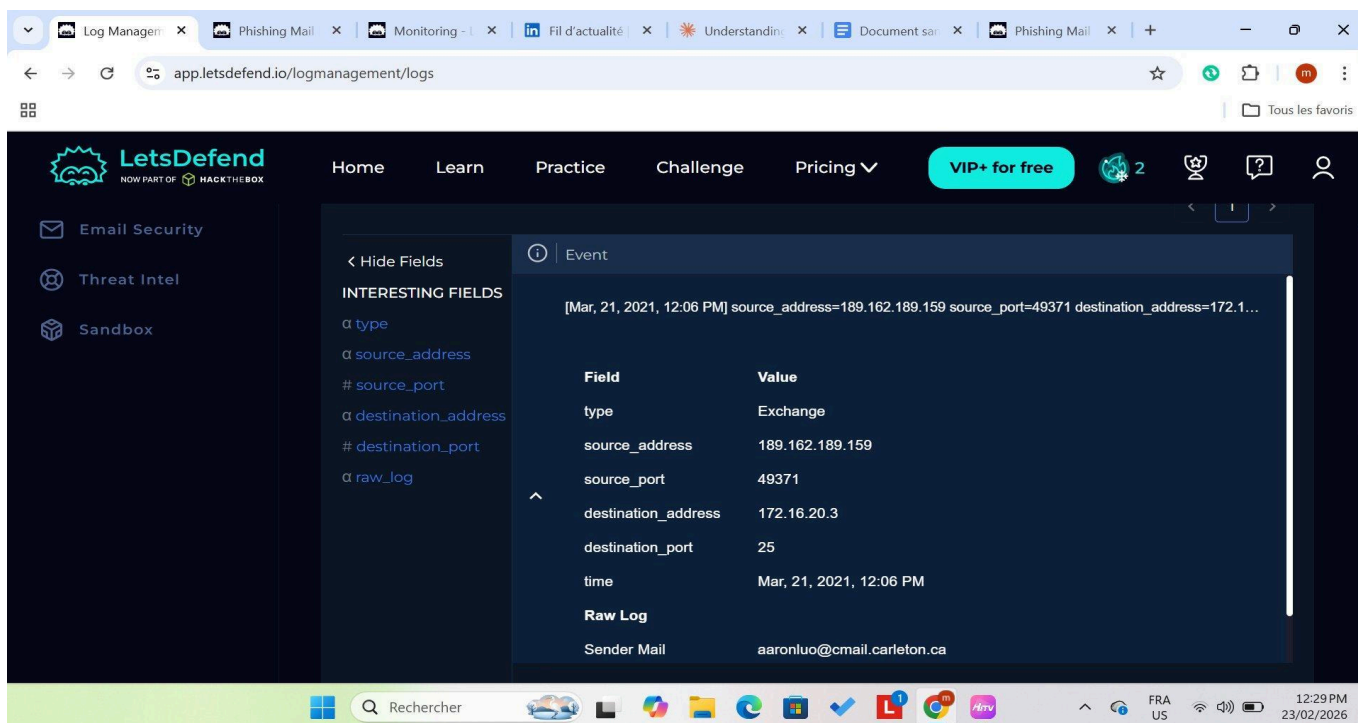
The following fields were extracted from the alert and the LetsDefend log management system:

Sent Date / Time	March 21, 2021, 12:26 PM
SMTP Address	189.162.189.159
Sender (From)	aaronluo@cmail.carleton.ca
Recipient (To)	mark@letsdefend.io
Email Subject	COVID19 Vaccine
Source Port	49371
Destination Port	25 (SMTP)
Log Entry Time	March 21, 2021, 12:06 PM
Log Type	Exchange

 Fig 1 – Alert Header Details (LetsDefend Alert Panel)

EventID :	82
Event Time :	Mar, 21, 2021, 12:26 PM
Rule :	SOC140 - Phishing Mail Detected - Suspicious Task Scheduler
Level :	Security Analyst
SMTP Address :	189.162.189.159
Source Address :	aaronluo@cmail.carleton.ca
Destination Address :	mark@letsdefend.io
E-mail Subject :	COVID19 Vaccine
Device Action :	Blocked

 **Fig 2 – Log Management Entry (SMTP Transaction Log)**



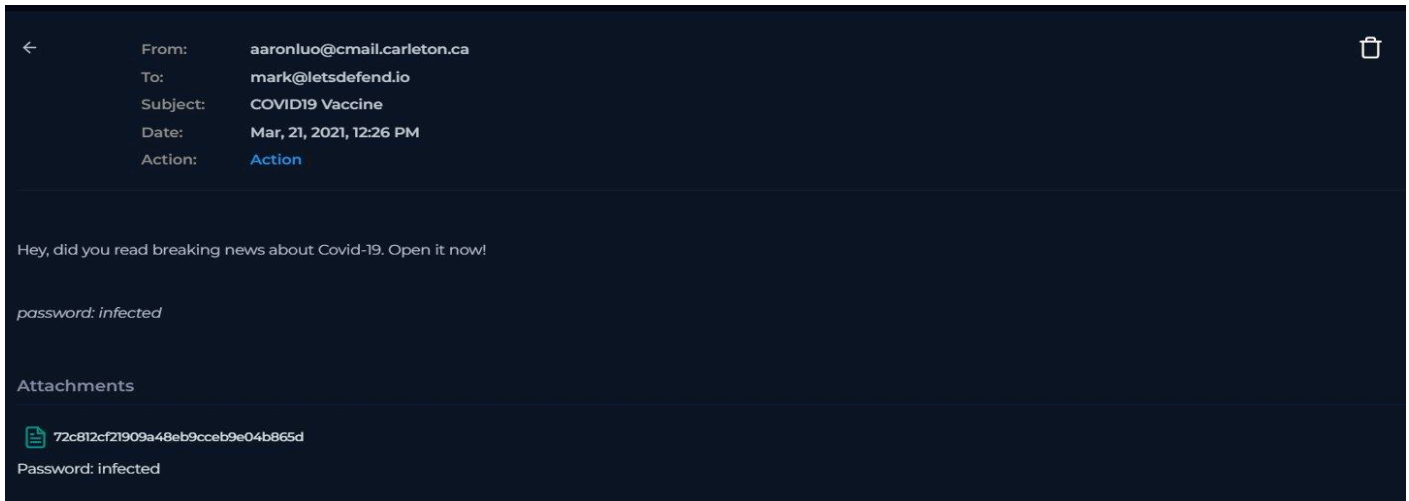
1.2 Email Body & Attachment

The email was accessed in the LetsDefend mailbox simulation. Content analysis revealed:

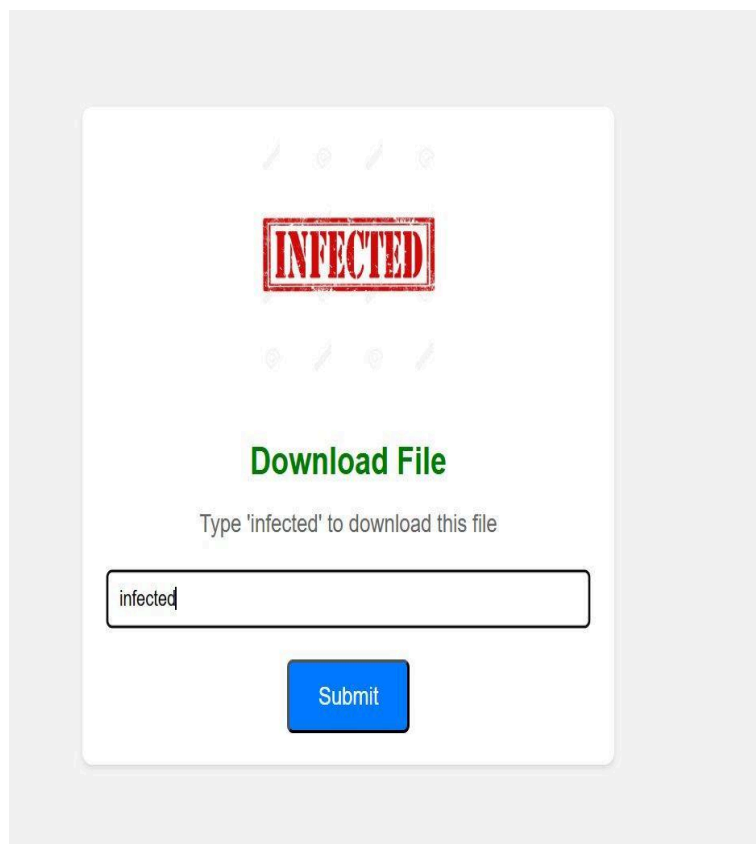
Body Text	"Hey, did you read breaking news about Covid-19. Open it now!"
Social Engineering	Urgency + pandemic fear lure
Password in Body	infected (plaintext embedded by attacker)
Attachment Name	72c812cf21909a48eb9cceb9e04b865d
Attachment Type	Password-protected ZIP archive
Attachment Password	infected
URLs / Links	Yes — embedded download URL (AWS S3)

 **AV Evasion:** The attacker deliberately embedded the ZIP password in the email body so the victim can open it — while simultaneously preventing email gateway scanners from inspecting the encrypted archive. This is a known technique to bypass secure email gateways.

 **Fig 3 – Email Body Content (LetsDefend Mailbox)**



 **Fig 4 – File Download Page (Password: 'infected')**



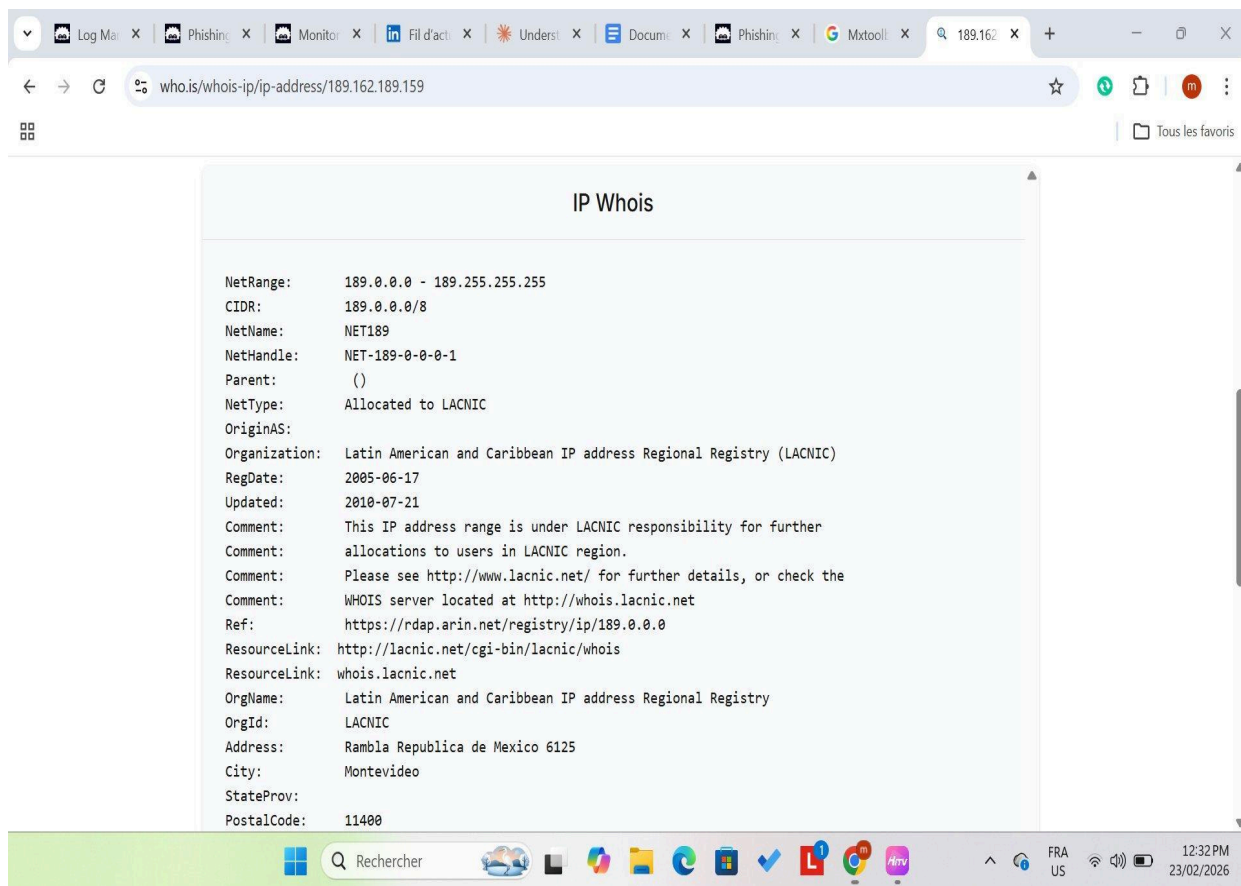
1.3 Sender IP Reputation — WHOIS Lookup (189.162.189.159)

IP Investigated	189.162.189.159
-----------------	-----------------

Registry	LACNIC (Latin American & Caribbean IP Registry)
City / Country	Montevideo, Uruguay
Claimed Sender Org	Carleton University — Ottawa, Canada
Geo Mismatch	YES — No evidence this IP belongs to Carleton infrastructure
Reverse DNS	No PTR record identified — IP not linked to any university hostname
Verdict	IP origin inconsistent with claimed sender — strong spoofing indicator

 **Geolocation Mismatch: No evidence suggests this IP (189.162.189.159) belongs to Carleton University infrastructure. Universities can use cloud or third-party mail relays, but this LACNIC-allocated IP in Uruguay shows no association with any Carleton-linked hostname or ASN. Combined with a non-existent sender domain and malicious payload, spoofing is the most credible conclusion.**

 **Fig 5 – WHOIS Lookup for 189.162.189.159 (LACNIC / Uruguay)**



1.4 SPF / DKIM / DMARC — Domain DNS Check (cmail.carleton.ca)

 Tool used: MXToolbox SuperTool — DNS Record lookup for cmail.carleton.ca

Domain Checked	cmail.carleton.ca
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DNS Record	✗ NOT FOUND — subdomain does not resolve
SPF Record	Not found on this subdomain (SPF is typically configured at root domain: carleton.ca)
DKIM Record	Not found on this subdomain — no cryptographic signing verifiable
DMARC Policy	Not found on this subdomain (DMARC usually sits at root domain level)
Domain Age	Cannot be determined — subdomain is non-resolvable
Key Finding	email.carleton.ca does not exist in DNS — it cannot send legitimate mail
Analyst Note	Absence of SPF/DMARC on a subdomain alone is not definitive proof of spoofing. The real indicators are: non-resolvable domain + IP geo mismatch + confirmed malicious payload

🚨 Primary proof of spoofing is NOT the absence of SPF/DMARC records on the subdomain (these may legitimately sit at the root domain carleton.ca). The real proof is: (1) email.carleton.ca does not exist in DNS, meaning no mail can legitimately originate from it, (2) the sending IP belongs to LACNIC/Uruguay with no Carleton association, and (3) the payload is confirmed malicious.

🔗 Fig 6 – MXToolbox DNS Lookup for email.carleton.ca (DNS Record Not Found)

SuperTool Beta9
DNS Lookup

a:email.carleton.ca
Find Problems

	Test	Result	
✗	DNS Record Published	DNS Record not found	More Info

[dns check](#)
[mx lookup](#)
[dmARC lookup](#)
[spf lookup](#)
[dns propagation](#)

Reported by ns2.carleton.ca on 2/23/2026 at 5:33:42 AM (UTC -6), [just for you.](#) [Transcript](#)

ABOUT THE SUPERTOOL

PHASE 2 — STATIC FILE ANALYSIS

2.1 Hash Extraction & VirusTotal File Hash Scan

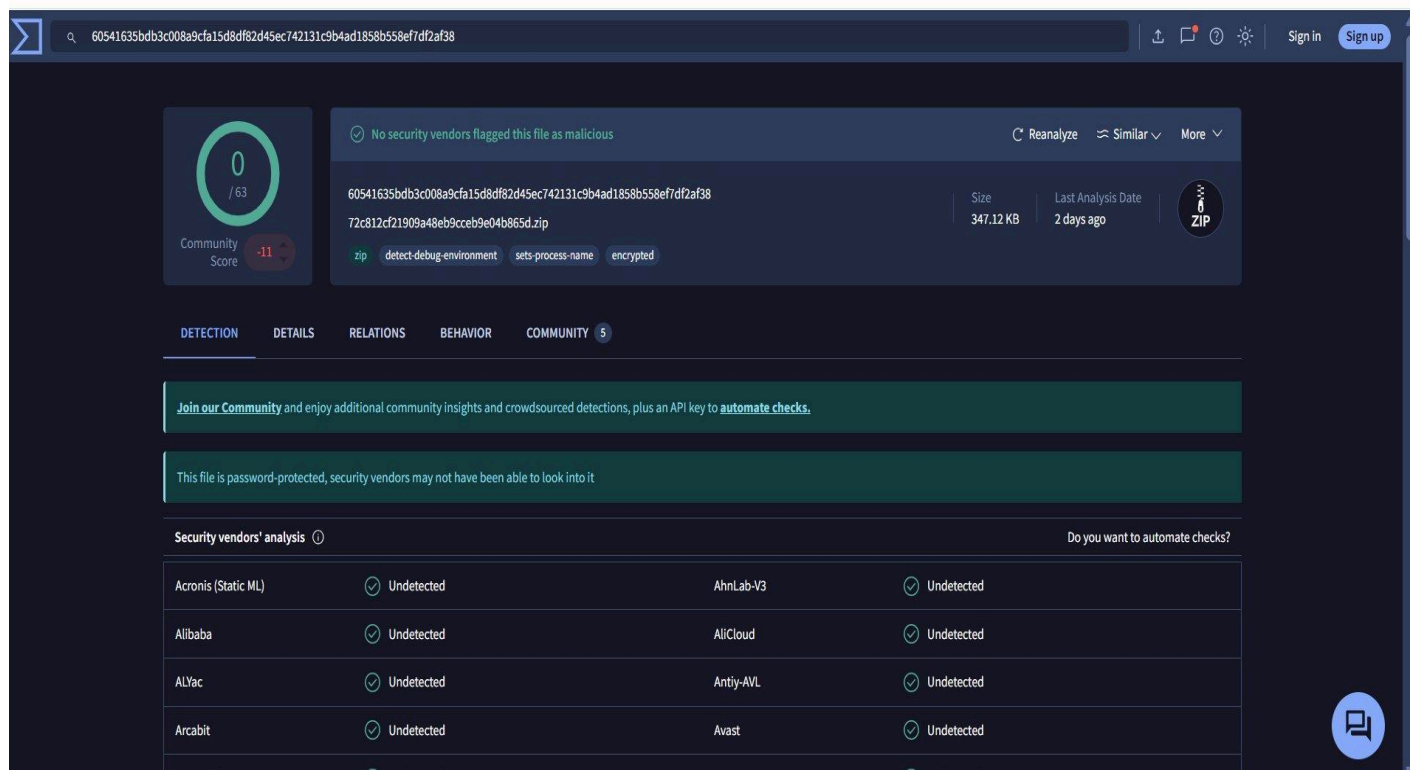
The SHA256 hash was extracted and submitted directly to VirusTotal for file reputation analysis:

Filename (MD5)	72c812cf21909a48eb9cceb9e04b865d
----------------	----------------------------------

File Size	347 KiB (347.12 KB per VirusTotal)
MIME Type	application/zip (password-protected / encrypted)
SHA256 Hash	60541635bdb3c008a9cfa15d8df82d45ec742131c9b4ad1858b558ef7df2af38
VT File Detection	0 / 63 — No AV engine flagged the file as malicious
VT Warning	"This file is password-protected, security vendors may not have been able to look into it"
VT Tags	zip, encrypted, detect-debug-environment, sets-process-name
VT Community Score	-11 (negative score — flagged as suspicious by the research community)
Last VT Analysis	2 days ago (from date of investigation — Feb 2026)
First Submitted	2021-03-22 03:38:18 UTC

 **AV Evasion Confirmed: The 0/63 detection is NOT a clean result — it is the evasion technique working as intended. VirusTotal itself acknowledges the file is password-protected and security vendors could not inspect it. The password 'infected' provided in the email body is the key the victim uses to open the archive, while AV engines remain unable to scan it. Sandbox detonation (Hybrid Analysis) bypasses this and returns a Threat Score of 100/100.**

 **Fig 7 – VirusTotal File Hash Scan (SHA256) — 0/63 Due to Password-Protected ZIP**



60541635bdb3c008a9cfa15d8df82d45ec742131c9b4ad1858b558ef7df2af38

Size: 347.12 KB | Last Analysis Date: 2 days ago | ZIP

Community Score: -11 / 63

Tags: zip, detect-debug-environment, sets-process-name, encrypted

DETECTION DETAILS RELATIONS BEHAVIOR COMMUNITY 5

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

This file is password-protected, security vendors may not have been able to look into it

Security vendors' analysis  Do you want to automate checks?

Acronis (Static ML)	Undetected	AhnLab-V3	Undetected
Alibaba	Undetected	AliCloud	Undetected
ALYac	Undetected	Antiy-AVL	Undetected
Arcabit	Undetected	Avast	Undetected
Avast-Mobile	Undetected	AVG	Undetected

2.2 URL Reputation — VirusTotal (Primary Download URL)

The AWS S3 download URL extracted from the email attachment was submitted to VirusTotal:

URL Scanned	https://files-ld.s3.us-east-2.amazonaws.com/72c812cf21909a48eb9cceb9e04b865d.zip
VT Score	8 / 94 security vendors flagged MALICIOUS
Flagged Engines	ADMINUSLabs, alphaMountain.ai, BitDefender, CyRadar
Threat Family	SILENTBUILDER — dropper/downloader (Conti ransomware subgroup)
Crowdsourced Intel	HIGH severity — Cluster25 threat intelligence
Content-Type	application/xml (payload disguised as data)
HTTP Status	403 Forbidden
Infrastructure	AWS S3 (cloud hosting abused for malware distribution)

🚨 **SILENTBUILDER:** This dropper is used by a subgroup of the Conti ransomware gang. The MSI payload within the ZIP disguises itself as a legitimate Notepad++ installer — a known Living-off-the-Land (LotL) delivery technique.

🔗 Fig 8 – VirusTotal Scan — Primary Download URL (8/94 Malicious, SILENTBUILDER)

https://files-ld.s3.us-east-2.amazonaws.com/72c812cf21909a48eb9cceb9e04b865d.zip?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Content-Sha256=UNSIGNED-PAYLOAD&X-Amz-Credential=AKIA6DVNZHVQL3M...

8 / 94 security vendors flagged this URL as malicious

https://files-ld.s3.us-east-2.amazonaws.com/72c812cf21909a48eb9cceb9e04b865d.zip?X-Amz-Algorithm=AWS4-HMAC-SHA256...
files-ld.s3.us-east-2.amazonaws.com

Status: 403
Content type: application/xml
La... a mo

application/xml

DETECTION DETAILS COMMUNITY

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Crowdsourced context ⓘ

HIGH 1 MEDIUM 0 LOW 0 INFO 0 SUCCESS 0

Activity related to SILENTBUILDER - according to source Cluster25 - 2 years ago
This DOMAIN is used by SILENTBUILDER. SilentBuilder is a dropper and downloader used by a subgroup of Conti. The MSI file downloaded appears to be a Notepad++ installer.

Security vendors' analysis ⓘ

Do you want to automate checks?

ADMINUSLabs	Malicious	alphaMountain.ai	Malicious
BitDefender	Malware	CyRadar	Malware

2.3 URL Reputation — VirusTotal (Secondary URL: download.cyberlearn.academy)

URL Scanned	https://download.cyberlearn.academy/download/download_file/
VT Score	9 / 94 security vendors flagged MALICIOUS
Flagged Engines	alphaMountain, BitDefender, CyRadar, Forcepoint, Fortinet, G-Data, Ionic, Sophos, VIPRE

Suspicious	Gridinsoft flagged as Suspicious
Content-Type	application/json; charset=utf-8
HTTP Status	404 (domain now offline)

 **Fig 9 – VirusTotal Scan — Secondary URL (9/94 Malicious)**

 Tous les



https://download.cyberlearn.academy/download/download_file/

Sign in Sig

9
/ 94
Community Score

9/94 security vendors flagged this URL as malicious

https://download.cyberlearn.academy/download/download_file/download.cyberlearn.academy

Status 404 Content type application/json; charset=utf-8 Last Analysis Date 22 days ago

application/json

DETECTION

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Security vendors' analysis

Do you want to automate checks?

alphaMountain.ai	Malicious	BitDefender	Malware
CyRadar	Malware	Forcepoint ThreatSeeker	Malicious
Fortinet	Malware	G-Data	Malware
Lionic	Malware	Sophos	Malware
VIPRE	Malware	Gridinsoft	Suspicious
Abusix	Clean	Acronis	Clean

https://www.virustotal.com/gui/url/b48ac0ea88a608646eae8e7ed51c8ad0d4217eda1299d67d286279711e2950ea/detection

2.4 URLhaus — Threat Database Lookup

Database	URLhaus by abuse.ch
Result	No records found
Interpretation	URL not yet submitted to URLhaus — does NOT indicate the URL is clean
Analyst Note	Confirmed malicious by VirusTotal, urlscan, and Hybrid Analysis — absence from URLhaus is not exonerating

 **Fig 10 – URLhaus Lookup (No Records — Not Yet Reported)**

In order to report a new malware URL to URLhaus, you need to login with your [abuse.ch](#) account

Browse Database

🔍 Search

🔗 URLs

⚙️ Payloads

Dateadded (UTC)	Malware URL	Status	Tags	Reporter
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Previous

Next

2.5 URLscan.io — Hosting Infrastructure Analysis

Domain Scanned	files-lid.s3.us-east-2.amazonaws.com
Resolved IP	3.5.128.229 (Amazon AWS — Columbus, US)
Google Safe Browsing	🚫 MALICIOUS
Times Scanned	1607 scans on urlscan.io
TLS Certificate	Amazon RSA 2048 M01 — issued July 11, 2025 (valid)
Infrastructure Abuse	Attackers host payload on legitimate AWS S3 to bypass IP reputation filters

🚫 Google Safe Browsing explicitly marks this AWS S3 domain as MALICIOUS — a high-confidence global signal. Abusing legitimate cloud infrastructure (AWS, GitHub, Google Drive) is a standard technique to evade domain-reputation-based blocklists.

📎 Fig 11 – URLscan.io Analysis (Google Safe Browsing: Malicious)

urlscan.io

[Home](#) [Search](#) [Live](#) [API](#) [Blog](#) [Docs](#) [Pricing](#) [Login](#)

files-ld.s3.us-east-2.amazonaws.com

3.5.128.229 [Public Scan](#)

[Lookup](#) [Go To](#) [Rescan](#) [Add Verdict](#) [Report](#)

URL: <https://files-ld.s3.us-east-2.amazonaws.com/72c812cf21909a48eb9cceb9e04b865d.zip?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Content-Sha256=UNSIG...>
Submission: On February 23 via manual (February 23rd 2026, 11:51:02 am UTC) from [DZ](#) — Scanned from [US](#)

[Summary](#) [HTTP](#) [Redirects](#) [Behaviour](#) [Indicators](#) [Similar](#) [DOM](#) [Content](#) [API](#) [Verdicts](#)

Summary

This website contacted 2 IPs in 1 countries across 1 domains to perform 2 HTTP transactions. The main IP is 3.5.128.229, located in Columbus, United States and belongs to AMAZON-02 - Amazon.com, Inc., US. The main domain is files-ld.s3.us-east-2.amazonaws.com. TLS certificate: Issued by Amazon RSA 2048 M01 on July 11th 2025. Valid for: a year.

files-ld.s3.us-east-2.amazonaws.com scanned 1607 times on urlscan.io [Show Scans](#) 1607

urlscan.io Verdict: No classification

Live information

Google Safe Browsing: Malicious for files-ld.s3.us-east-2.amazonaws.com
Current DNS A record: 3.5.132.216 (AS16509 - AMAZON-02 - Amazon.com, Inc., US)

Domain & IP information

[IP/ASNs](#) [IP Detail](#) [Domains](#) [Domain Tree](#) [Links](#) [Certs](#) [Frames](#)

	IP Address	AS Autonomous System
2	3.5.128.229	16509 (AMAZON-02 - Amazon.com)
2	2	

Screenshot

[Live screenshot](#) [Full Image](#)

Detected technologies

Amazon S3 (CDN) [Expand](#)

Page Statistics

2100 %0 %11

RequestsHTTPSIPv6DomainsSubdomains

211 kB1 kB0

IPsCountriesTransferSizeCookies

PHASE 3 — DYNAMIC ANALYSIS (SANDBOX DETONATION)

3.1 Why Dynamic Analysis Was Performed

Although the email was blocked at the gateway (True Positive confirmed), dynamic sandbox analysis was conducted to fully understand attacker TTPs, payload behavior, and to extract IOCs for threat intelligence — standard practice for analyst upskilling and threat documentation.

3.2 Hybrid Analysis — Falcon Sandbox Results

Submission Name	72c812cf21909a48eb9cceb9e04b865d.zip
SHA256	60541635bdb3c008a9cfa15d8df82d45ec742131c9b4ad1858b558ef7df2af38
File Size	347 KiB
MIME Type	application/zip (password-protected)
Threat Score	100 / 100 — MAXIMUM THREAT LEVEL
Sandbox Verdict	MALICIOUS

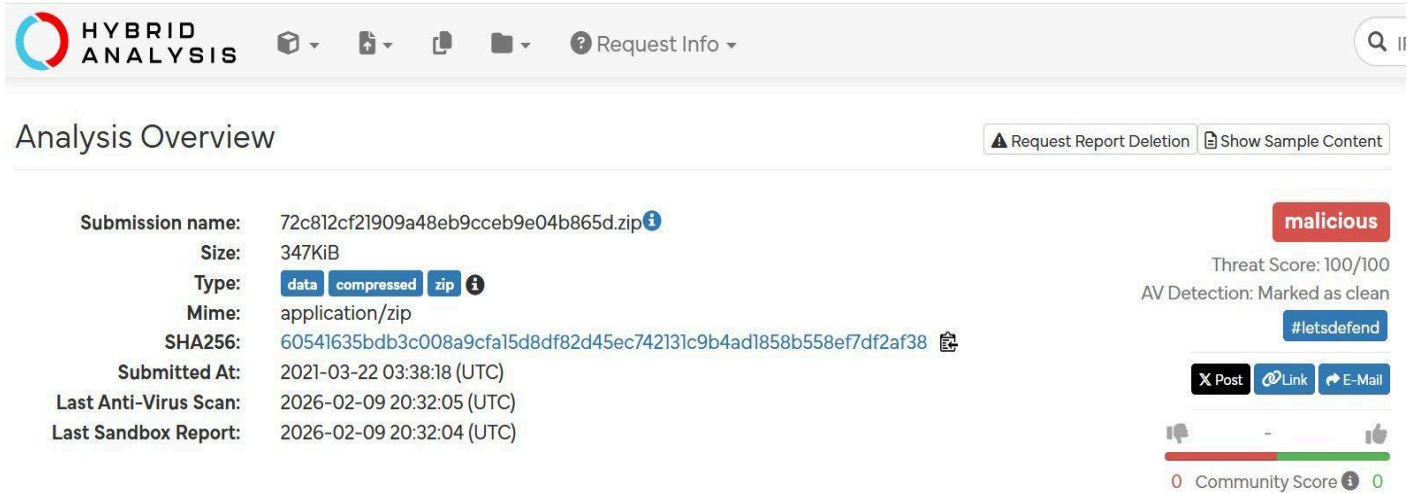
AV Detection	Marked as Clean (AV cannot scan password-protected ZIP — evasion confirmed)
Bundled Files	1 malicious file extracted within ZIP
Associated URL	hxxps://files-lf.s3.us-east-2.amazonaws.com/72c812cf21909a48eb9cc9eb9e04b865d.zip

Sandbox Detonation Results by Environment:

- Windows 7 32-bit → Successfully detonated — malicious behavior confirmed
- Windows 7 64-bit → Successfully detonated — malicious behavior confirmed
- Windows 10 64-bit → Successfully detonated — malicious behavior confirmed
- Windows 11 64-bit → Successfully detonated — malicious behavior confirmed
- Linux (Ubuntu 20.04 / 16.04) → Extraction error — payload is Windows-specific (MSI)

The AV scan reported 'Clean' despite a 100/100 threat score. This is not a contradiction — AV cannot inspect the encrypted ZIP contents without the password. The sandbox bypasses this by detonating the file in a controlled Windows environment, which triggers the payload and reveals its true behavior.

 **Fig 12 – Hybrid Analysis Report (Threat Score: 100/100 — Malicious)**



PHASE 4 — INDICATORS OF COMPROMISE (IOCs)

Type	Value	Source / Notes
IP Address	189.162.189.159	WHOIS — LACNIC / Uruguay (geo mismatch)
Email	aaronluo@cmail.carleton.ca	MXToolbox — domain has no DNS record
Domain	cmail.carleton.ca	Spoofed subdomain — DNS not found
URL (Primary)	files-lf.s3.us-east-2.amazonaws.com/72c812..zip	VT 8/94 + Google Safe Browsing Malicious

URL (Secondary)	download.cyberlearn.academy/download/download_file/	VT 9/94 Malware (9 engines)
Hash MD5	72c812cf21909a48eb9cceb9e04b865d	Hybrid Analysis — filename / MD5
Hash SHA256	60541635bdb3c008a9cfa15d8df82d45ec742131c9b4ad1858b558ef7df2af38	Hybrid Analysis — Threat Score 100/100
Subject	COVID19 Vaccine	Social engineering lure
Password	infected (ZIP password embedded in email body)	AV evasion technique
Threat Actor	SILENTBUILDER (Conti ransomware subgroup)	VirusTotal Cluster25 crowdsourced intel

PHASE 5 — MITRE ATT&CK® MAPPING

The following ATT&CK techniques were identified based on observable evidence from this investigation:

Technique ID	Name	Tactic	Evidence in This Case
T1566.001	Phishing: Spearphishing Attachment	Initial Access	Email with malicious ZIP attachment sent to mark@letsdefend.io
T1036	Masquerading	Defense Evasion	Sender spoofed as Carleton University student; payload disguises itself as Notepad++ MSI installer
T1027	Obfuscated Files / Info	Defense Evasion	ZIP protected with password 'infected' — prevents AV/email gateway from scanning the archive
T1204.002	User Execution: Malicious File	Execution	Victim prompted to type password 'infected' and submit to download and open the ZIP
T1105	Ingress Tool Transfer	Command & Control	Payload downloaded from AWS S3 (files-ld.s3.us-east-2.amazonaws.com)
T1053	Scheduled Task / Job	Persistence / Execution	Rule name SOC140 references 'Suspicious Task Scheduler' — consistent with SILENTBUILDER dropper behavior
T1071.001	Application Layer Protocol: Web	Command & Control	SILENTBUILDER communicates over HTTPS; secondary C2 URL: download.cyberlearn.academy

PHASE 6 — RISK ASSESSMENT

Risk Level

 HIGH

Likelihood	HIGH — Multi-stage attack confirmed: spoofed domain + geo-mismatched SMTP + social engineering + malware delivery
Impact	CRITICAL — SILENTBUILDER is a dropper linked to Conti ransomware group; successful execution could lead to full domain compromise
Business Impact	Potential ransomware deployment, data exfiltration, lateral movement, and domain-wide encryption
Affected User	mark@letsdefend.io (targeted employee — email blocked before delivery)
Blast Radius	If executed: lateral movement risk to all internal systems reachable from compromised endpoint
Detection Outcome	Email BLOCKED by gateway — no user interaction occurred — containment successful ✓
Residual Risk	LOW post-blocking — IP and domain must be permanently blacklisted to prevent re-delivery attempt

PHASE 7 — RECOMMENDED ACTIONS & CLOSURE

1	Block IP 189.162.189.159 at firewall and email gateway — permanently.
2	Blacklist domain cmail.carleton.ca at email security gateway.
3	Block AWS S3 URL: files-ld.s3.us-east-2.amazonaws.com at web proxy.
4	Block secondary URL: download.cyberlearn.academy at web proxy.
5	Add SHA256 hash 60541635bdb3c008...f38 to EDR deny list.
6	Confirm mark@letsdefend.io received no other variant of this email.
7	Submit SHA256, URLs, and IPs to URLhaus and internal threat intel platform.
8	Escalate to Tier 2 / Threat Hunting — search environment for SILENTBUILDER/Conti TTPs.
9	Close alert as TRUE POSITIVE. Document all IOCs in SIEM and case management system.